Jaemoon Lee

Curriculum Vitae

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EDUCATION

Seoul National University *Mar. 2, 2025–present*
Ph.D. in Mathematical Sciences
Advisor: Prof. Seung-Yeal Ha.

Seoul National University *Mar. 2, 2019–Feb. 26, 2025*
B.Sc. in Mathematical Sciences and Artificial Intelligence

RESEARCH INTERESTS

Collective dynamics, kinetic equations, flocking models, synchronization, interacting particle systems, and emergent behavior.

FELLOWSHIPS AND AWARDS

- Presidential Scholarship for Graduate Students, Korea Student Aid Foundation (Spring 2025–present).
- Kwanjeong Educational Foundation Scholarship, Kwanjeong Educational Foundation (Spring 2025–present).
- Summa cum laude, Seoul National University (2025).
- Dean's List, College of Natural Sciences, Seoul National University (Fall 2023, Fall 2024).
- Scholarship for Academic Excellence, Seoul National University (Spring 2020, Spring 2021, Fall 2023, Spring 2024).

PUBLICATIONS

Journal Articles and Accepted Papers

1. S.-Y. Ha, **J. Lee**, Q. Xiao, and F. Zeng, A priori estimates on the weak flocking of the Cucker–Smale–Fokker–Planck equation. *Applied Mathematics Letters*, **176**, (2026), 109861.* Link
2. S.-Y. Ha, C. Jin, **J. Lee**, and Y. Zhang, Robustness of multi-cluster flocking and uniform-in-time transition for the discrete Motsch–Tadmor model. *Communications on Pure and Applied Analysis*, **34**, (2026), 95–121. Link

Conference Publications

1. G. Cho, **J. Lee**, J. Im, S. Lee, J. Lee, and S. Kim, AMPED: Adaptive multi-objective projection for balancing exploration and skill diversification. *International Conference on Learning Representations* (*ICLR*), **14**, (2026).† Link

Preprints and Submitted Manuscripts

1. R. Cheng, S.-Y. Ha, **J. Lee**, and Z. Wang, On the kinetic equation arising from the large-scale limit of the Cucker–Smale model. *arXiv*: 2602.14685. Submitted. Link
2. Q. Deng, S.-Y. Ha, and **J. Lee**, Particle, kinetic and hydrodynamic models for sea ice floes. Part II: Rotating floes with nonlinear contact forces. *arXiv*: 2602.21972. Submitted. Link
3. S.-Y. Ha, **J. Lee**, and F. Zeng, Formation of stochastic nematic alignment for the discrete-time Justh–Krishnaprasad model. Submitted.
4. S.-Y. Ha, C. Lee, E. Lee, **J. Lee**, and S.-Y. Ryoo, Emergent behaviors of Winfree oscillators on special orthogonal group. *arXiv*: 2604.24229. Submitted. Link

* Corresponding author. † First author.

TALKS

- Short presentation, Professor Seung-Yeal Ha's HYKE group seminar, Seoul National University, 2023–2026.
- Short presentation, Symposium on Collective Behavior in Nature and Science, Harbin Institute of Technology, School of Mathematics, Aug. 2, 2025.
- Short presentation, Workshop on Collective Dynamics of Interacting Particle Systems, Jilin University, School of Mathematics, Aug. 4, 2025.
- Poster, Summer School on Mathematical Neuroscience, Northwestern University, Aug. 25, 2025.
- Short presentation, Dalian International Conference on Mathematics, Dalian University of Technology, School of Mathematics, Sept. 22, 2025.
- Short presentation, Special Lectures, Yau Mathematical Sciences Center, Jingzhai, Tsinghua University, Mar. 26, 2026.
- Short presentation, Xêmina Phòng Toán rời rạc, Xác suát Thóng kê, Institute of Mathematics, Vietnam Academy of Science and Technology, Apr. 16, 2026.

VISITING

- Shandong University, School of Mathematics, Shandong, China, Nov. 5–8, 2025.
- Peking University, Beijing International Center for Mathematical Research, Beijing, China, Dec. 15, 2025–Jan. 13, 2026.
- Dalian University of Technology, Dalian, China, Apr. 19–24, 2026.

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